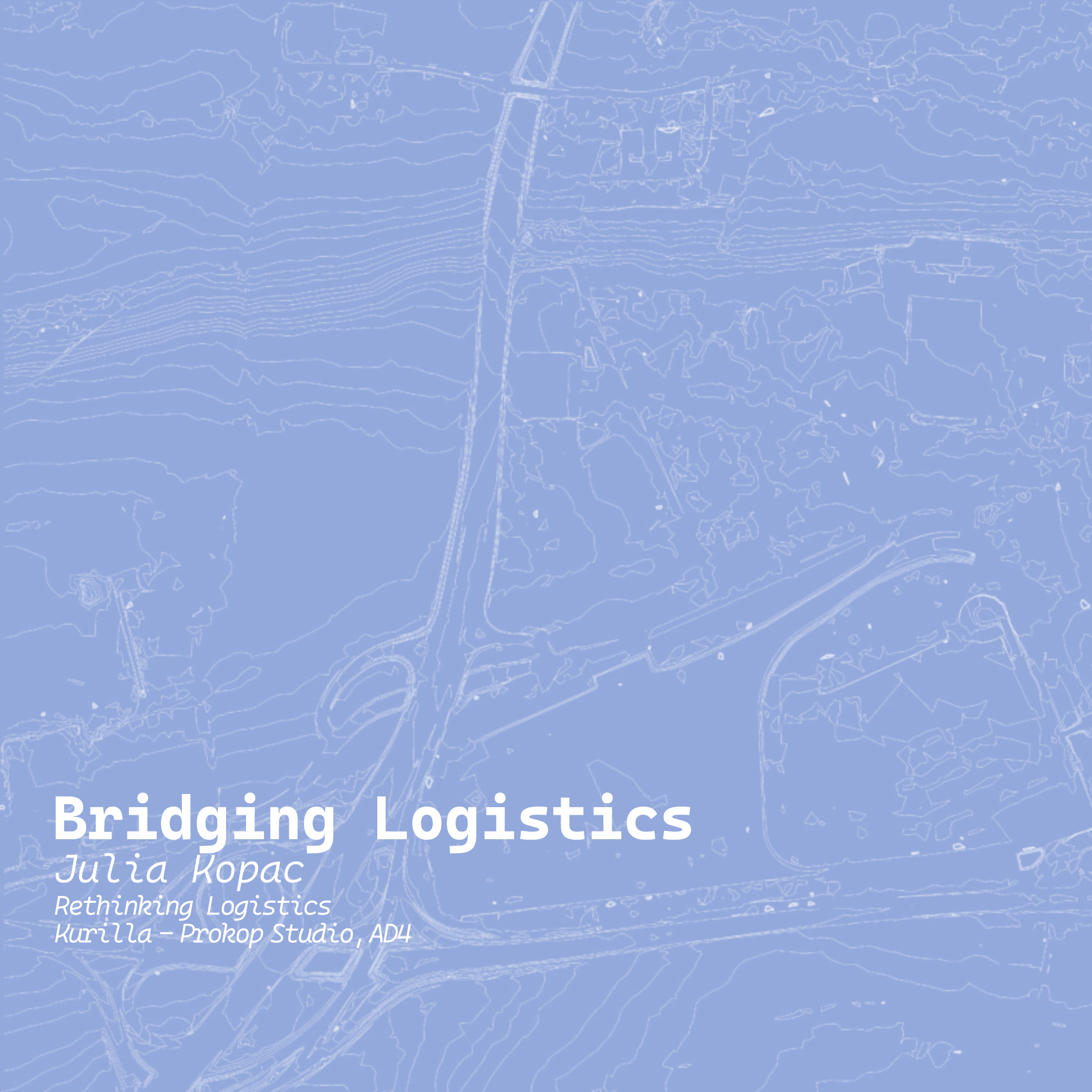




# Bridging Logistics

*Julia Kopac*

*Rethinking Logistics*

*Kurilla – Prokop Studio, AD4*

## Studio Brief

This project began with Rethinking Logistics, a studio exploring how warehouses and logistics facilities can be transformed from environmental burdens into regenerative urban assets. Developed in collaboration with Ramboll, Henning Larsen, Panattoni, and ULI Europe, the brief challenged rethinking industrial architecture through sustainable, evidence-based design strategies.

The studio focused on issues such as carbon footprint, biodiversity, stormwater management, and urban integration, encouraging proposals that balance environmental performance with real-world industrial and regulatory conditions. Through interdisciplinary collaboration and data-driven design methods, the project investigates how logistics architecture can benefit the future of a city.

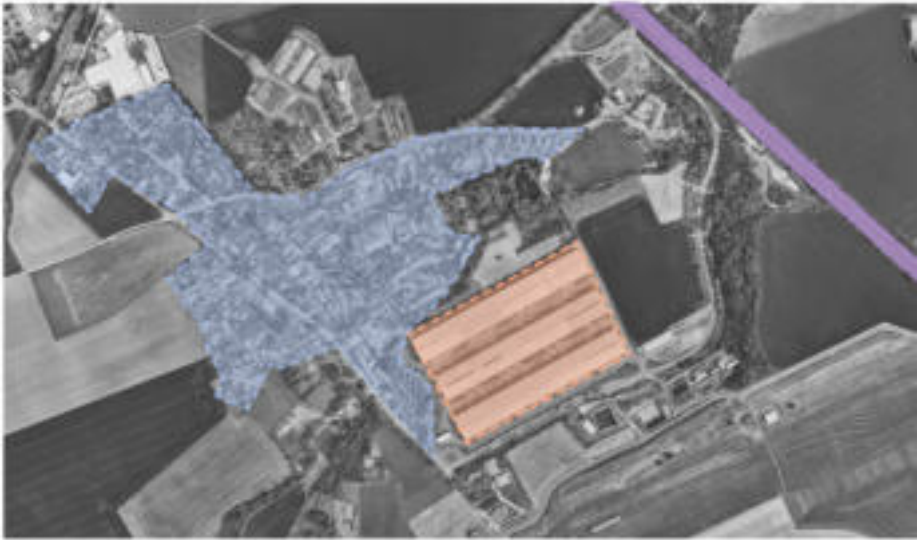
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# Analysis

In the beginning of the project the topic was very broad. In order to gain familiarity with the subject, the design process began with analysing exhisting logistics sites in Prague.

The warehouse areas on the border of the city all shared a common issue. All were extremely close to residential areas, divided by or next to a highway, and therefore restricted people from greenery.



CTPark Prague Airport, Prague



CTPark Prague North, Prague



Caravan Metropol, Klíčany-Vodochody



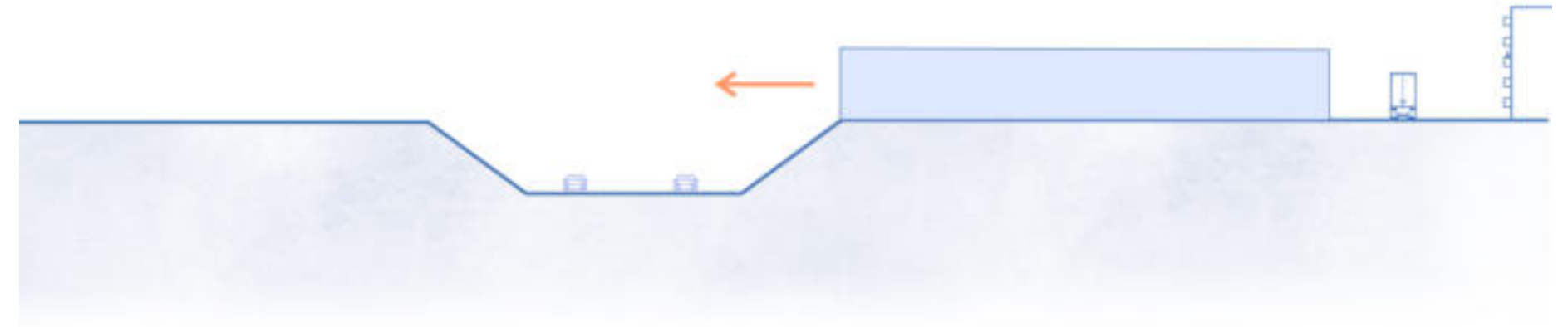
Business Park Zlicin, Prague

## Design Principle

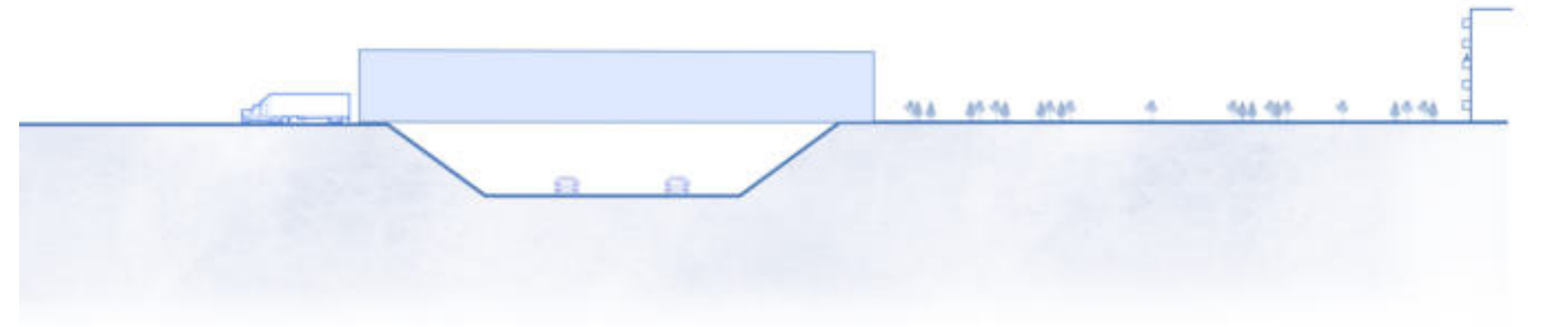
From the initial analysis followed by many iterations, the project demonstrates a basic design proposal that would solve the issues in the area.

Most warehouses are in the proximity of highways, built lower in the ground to help with noise levels. Moving the warehouses over highways solves the 3 main issues derived from the analysis.

It creates a more comfortable environment for residents, allows space for greenery, and blocks the high noise levels.



*Initial layout of a warehouse and residential area.*



*After proposed change, creating a more comfortable environment for all.*

# Site Study

After finding the concept, the analysis continued on one of the previously mentioned sites. This in-depth look at the site brought attention to more issues that the design should target.

The extensive examination in Business Park Zlicin showed that the design can aid with much more than was initially presumed.





Noise Pollution

As suggested previously, noise is a big issue in these areas. On this specific site, near the highway intersection, the noise can reach 75 dB. Noise protection is present only a few kilometers above the site and not near it.



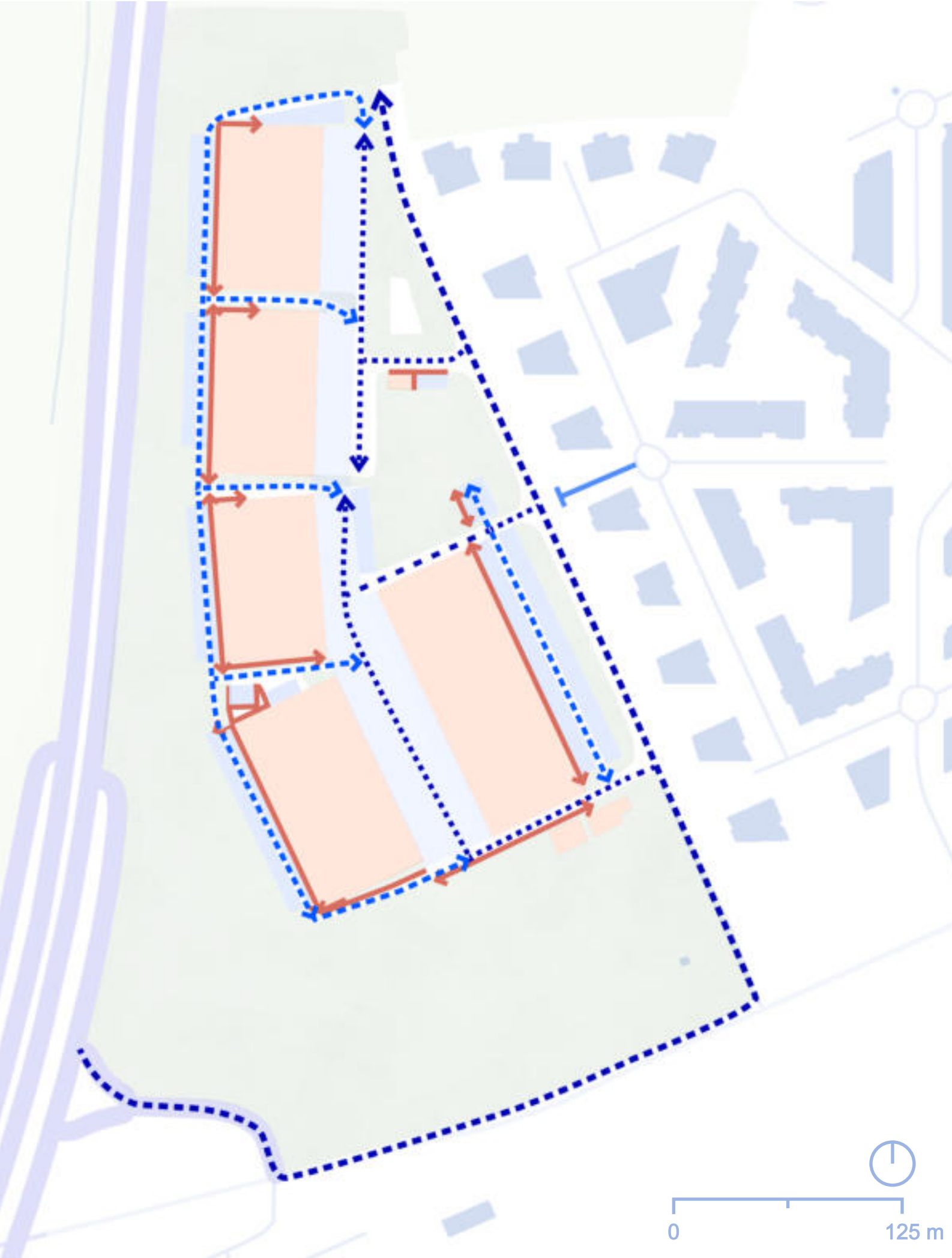
Connectivity and Safety

Another issue to take into account is the transportation. Only busses are available directly in the area since it is on the edge of the city, and even then, the stops are still at an inconvenient distance and in proximity to unsafe roads for workers.



Greenery : Artificial Materials

On the site, the logistics area takes up most of the viable green space, which could be better used for residents' leisure instead of being covered by concrete and parking space.



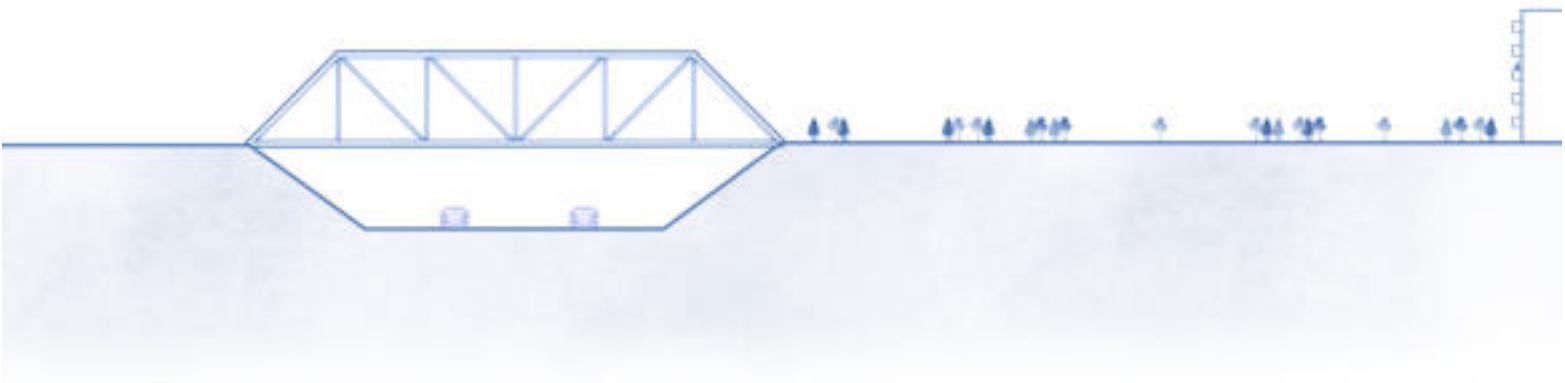
Within the premises, the increased truck and car routes increase pedestrian and worker risk. Additionally, the residential area is blocked by an incomplete lane, which does not connect to another road and is only used for parking. Lastly, the end of the main road leading into the site is used as parking for long haul truck drivers, creating an uncomfortable environment on the side of the road.

# Structure Examination

## Design

Following the research behind what the design should solve, the next steps dove into researching actual structures, and creating a modular system that would cater to all the problems.

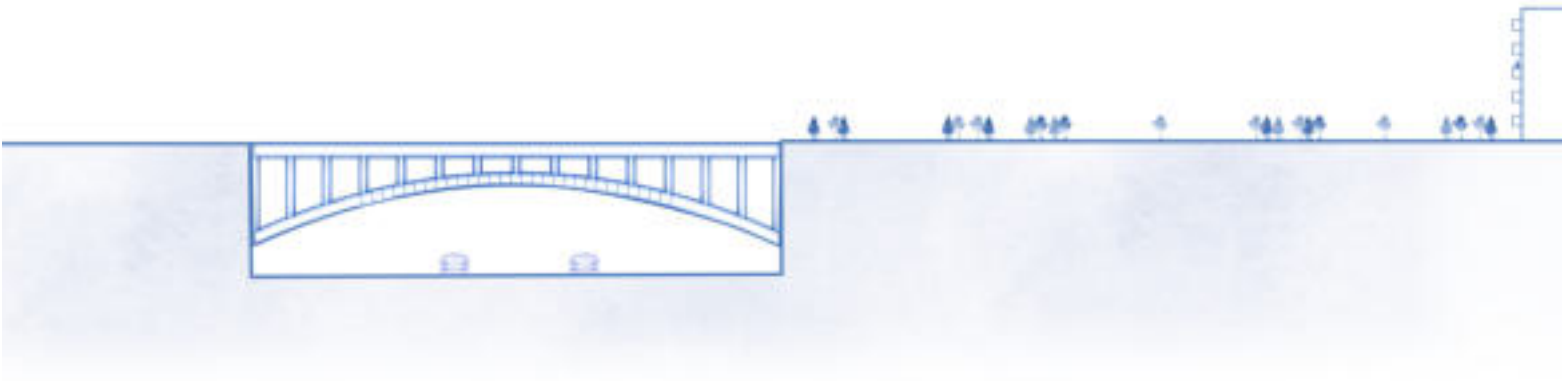
The structure needs to withstand the heavy loading from the warehouses, but still be light weight enough to be efficient over a highway.



Pratt truss

**Pros:** Easier to manufacture and assemble and no horizontal thrust needed, making it cheaper to seat on standard soil.

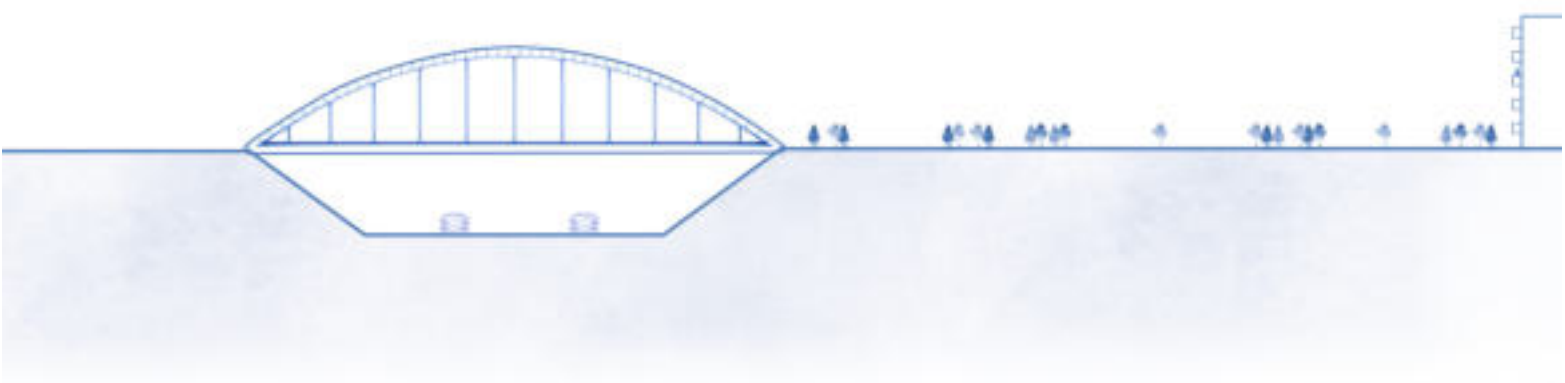
**Cons:**Structure requires a lot of steel which adds to the dead load and vibration have a significant effect on trusses.



Arch

**Pros:** Provides the most open space for warehouses above. Naturally very stiff and handles the heavy weight of a building well keeping everything in compression.

**Cons:**Requires expensive deep foundations since the structure has no tie.

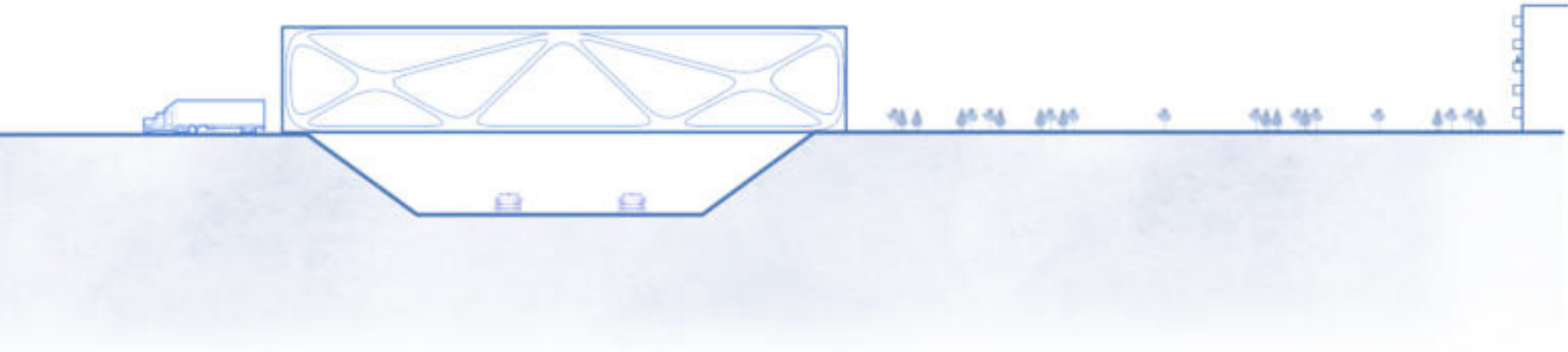


Suspended Arch

**Pros:** The tension tie offers stiffness, which allows for simpler foundations and minimizes foundations. The geometry also to handle high loads with less material weight.

**Cons:** Tension ties need maintenance, since if its damaged the structure could fail.

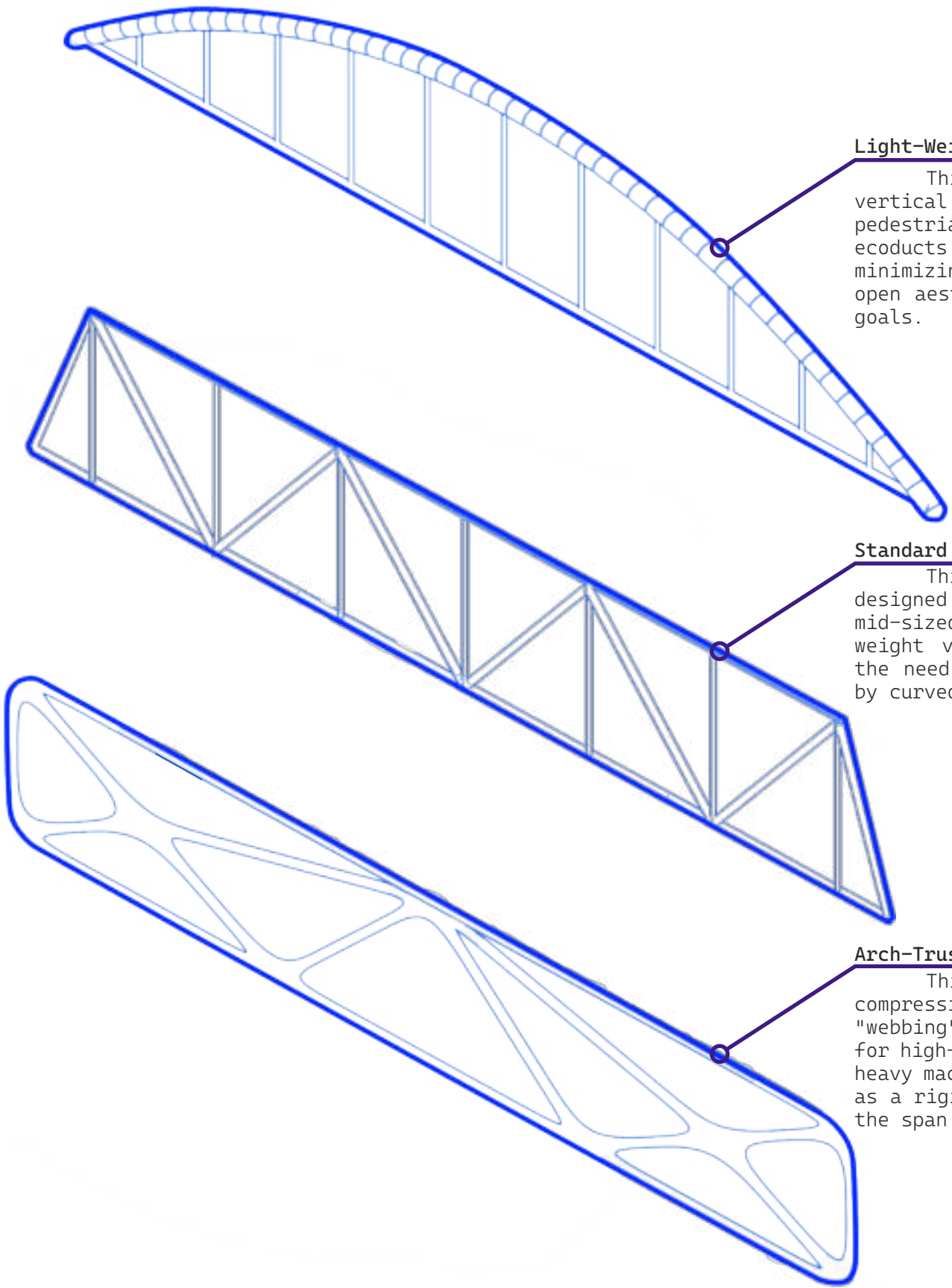
# Optimised structure



The hybrid arch-truss structure is the most optimized version to use since warehouses are not just a dead load but involve dynamic loads.

The arch component acts load-bearing. It handles the massive weight of the warehouse, while the truss component acts as a secondary stabilizer. In addition, the diagonal webbing resists shear forces and prevents the structure from swaying or racking.

# Structure variables

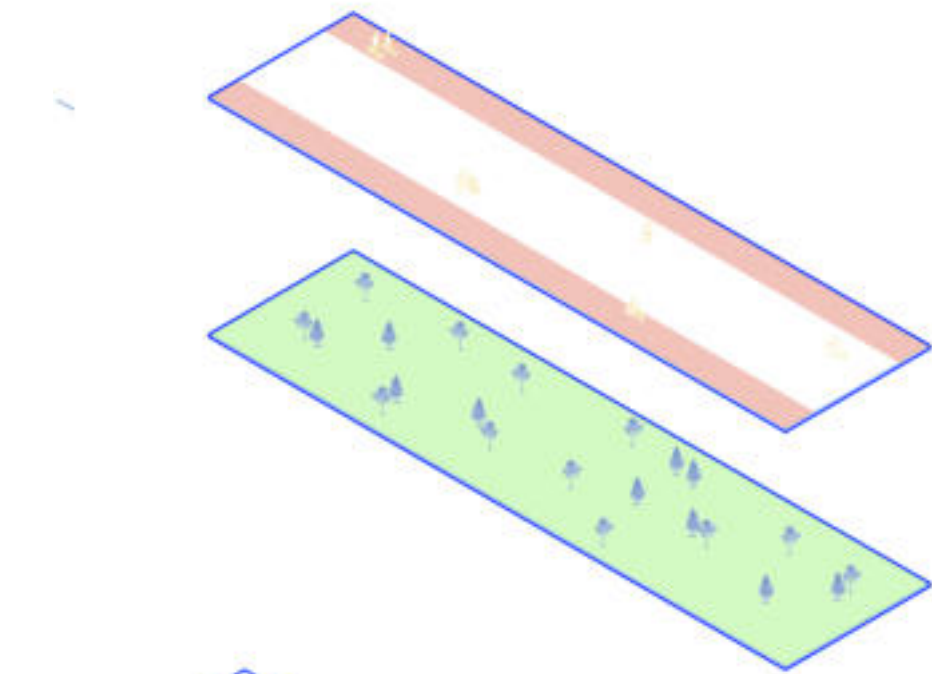


**Light-Weight Tied-Arch Module**  
This module utilizes a slender arch and thin vertical ties to support light "live loads" like pedestrian walking paths, bike lanes, or green ecoducts. It is best used for community crossings where minimizing material weight and maintaining a high, open aesthetic for the highway below are the primary goals.

**Standard Pratt Truss Module**  
This module is a self-contained, rigid system designed for moderate warehouse loads. It is ideal for mid-sized multi-tenant storage because it transfers weight vertically to standard foundations, avoiding the need for the massive concrete abutments required by curved structures.

**Arch-Truss Hybrid Module**  
This heavy-duty module integrates the compression strength of a deep arch with truss "webbing" in the spandrel corners. It is specifically for high-capacity warehouses that carry massive cargo, heavy machinery, and vibrating forklift traffic, acting as a rigid "hollow beam" to maintain stability across the span.

# Variability

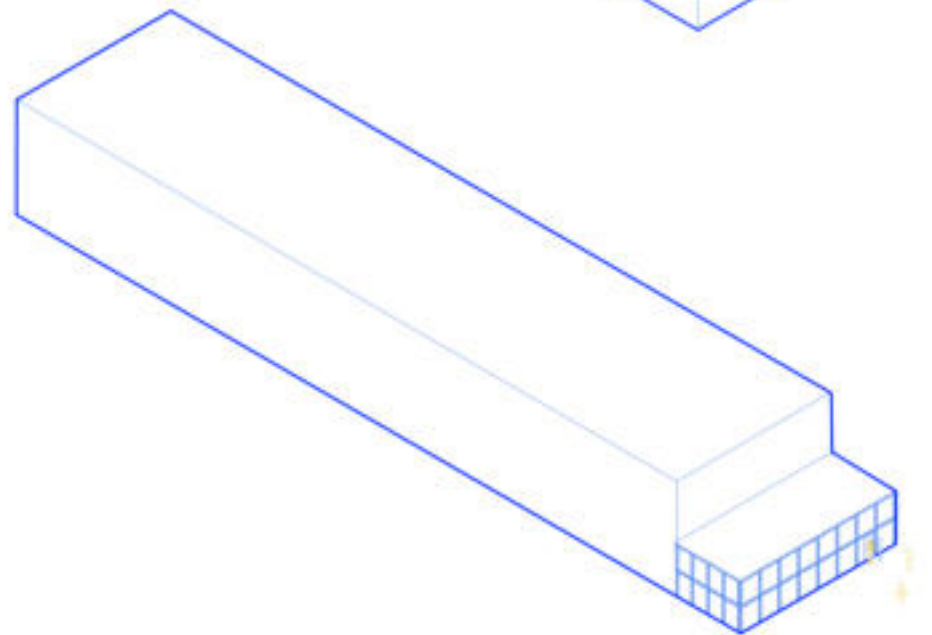
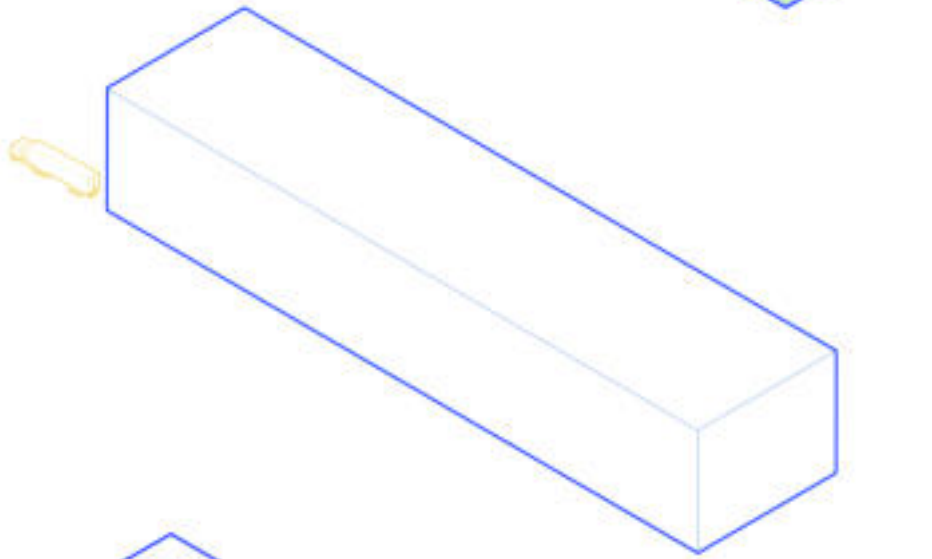


## Crossing

The crossing module provides an area for pedestrians and animals to safely cross without the obstacle of a highway and roads busy with trucks. One decking option is with a pedestrian walkway and bicycle lanes, the other, an ecoduct for near by animals to safely cross 4 lane highways to nearby reservoirs.

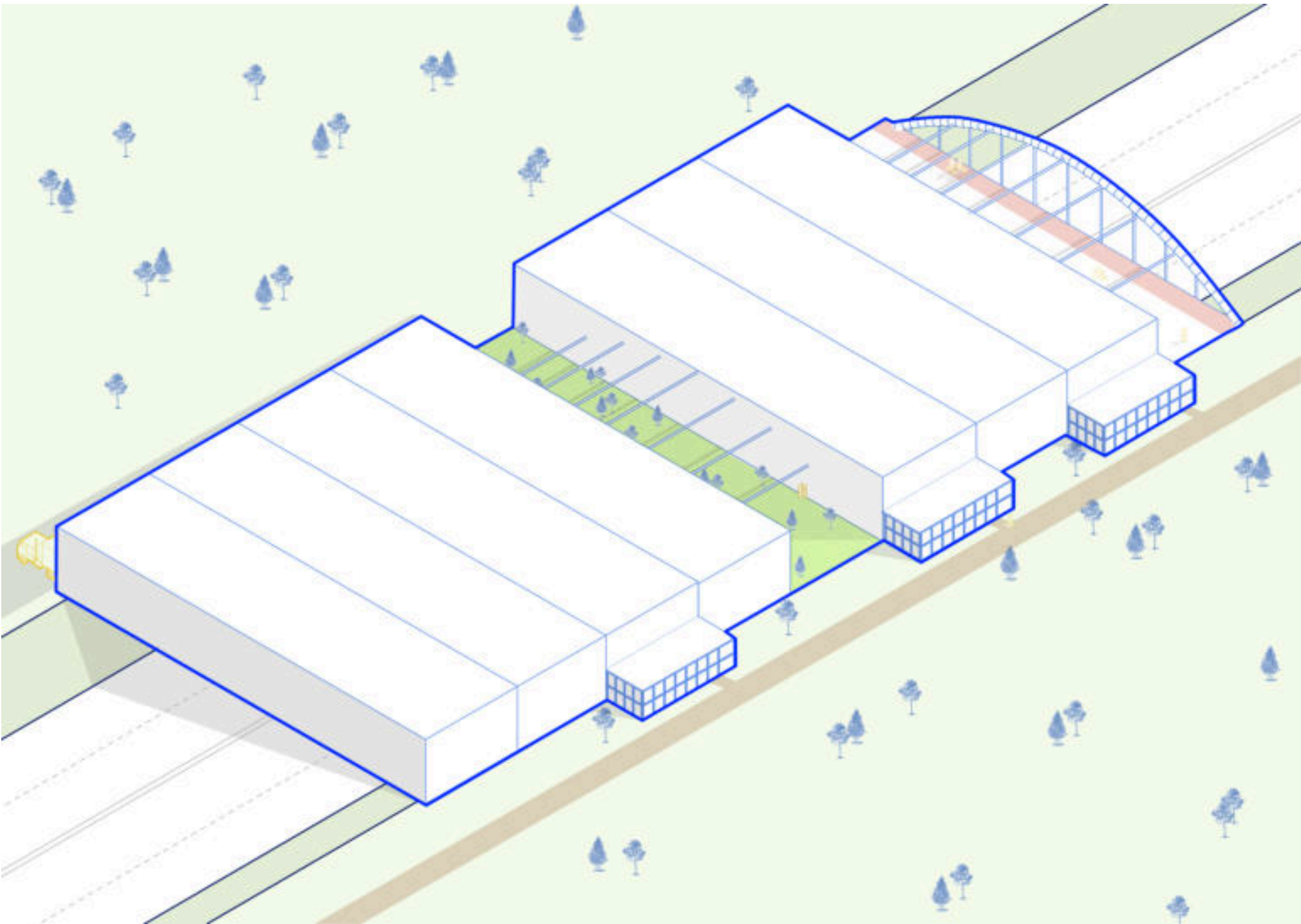
## Warehouse

The warehouse module is made to replace the standard warehouse space. The structure does not block the normal workings of the warehouse and it can be extended by any amount of modules. Additionally the truck traffic is placed on the other side of the highway creating safety for residents.



## Retail warehouse

The warehouse and retail model gives access for people to the warehouses directly. It still keeps the heavy traffic blocked from pedestrians. This version, if paired with a walkway module, can open up the retail spaces directly on the bridge.



Possible configuration of modules adapted on a potential site.

# Application on site



Previously mentioned site in Zlicin, Prague.



Proposal of possible changes that could happen after new design is applied.

Logistics hub

Residential area

Highway

# KPI

The design decisions were also heavily based on key performance indicators encouraging the best possible solution out of the design. The KPIs chosen heavily depended on my concept, focusing on what is important. This graph compares the usual warehouse designs (grey) with the new solution (blue).

## Uninterrupted Greenery

The area of continuous, undisturbed natural vegetation or green space on the site in m²



## Acoustics

The sound level measured at the edge of the highway in dB .



## Walkability

Score from 1-5 out of multiple categories: sidewalk continuity, Crossing Safety, Separation form vehicles, Perceived Safety and averaging the results.



## Area of Built Materials

Represents the total surface area covered by constructed materials (concrete floors, parking, roads) in % compared to the whole site.



## C02 Emissions to build

Captures the total embodied greenhouse gas emissions from all materials, transportation, and construction activities (embodied carbon) in kgCO₂e/m².



## Cost to build

Represents the total capital expenditure for construction and fit-out, normalized per square meter of floor area in eur/m²





Where everyday life and the warehouses meet in harmony.

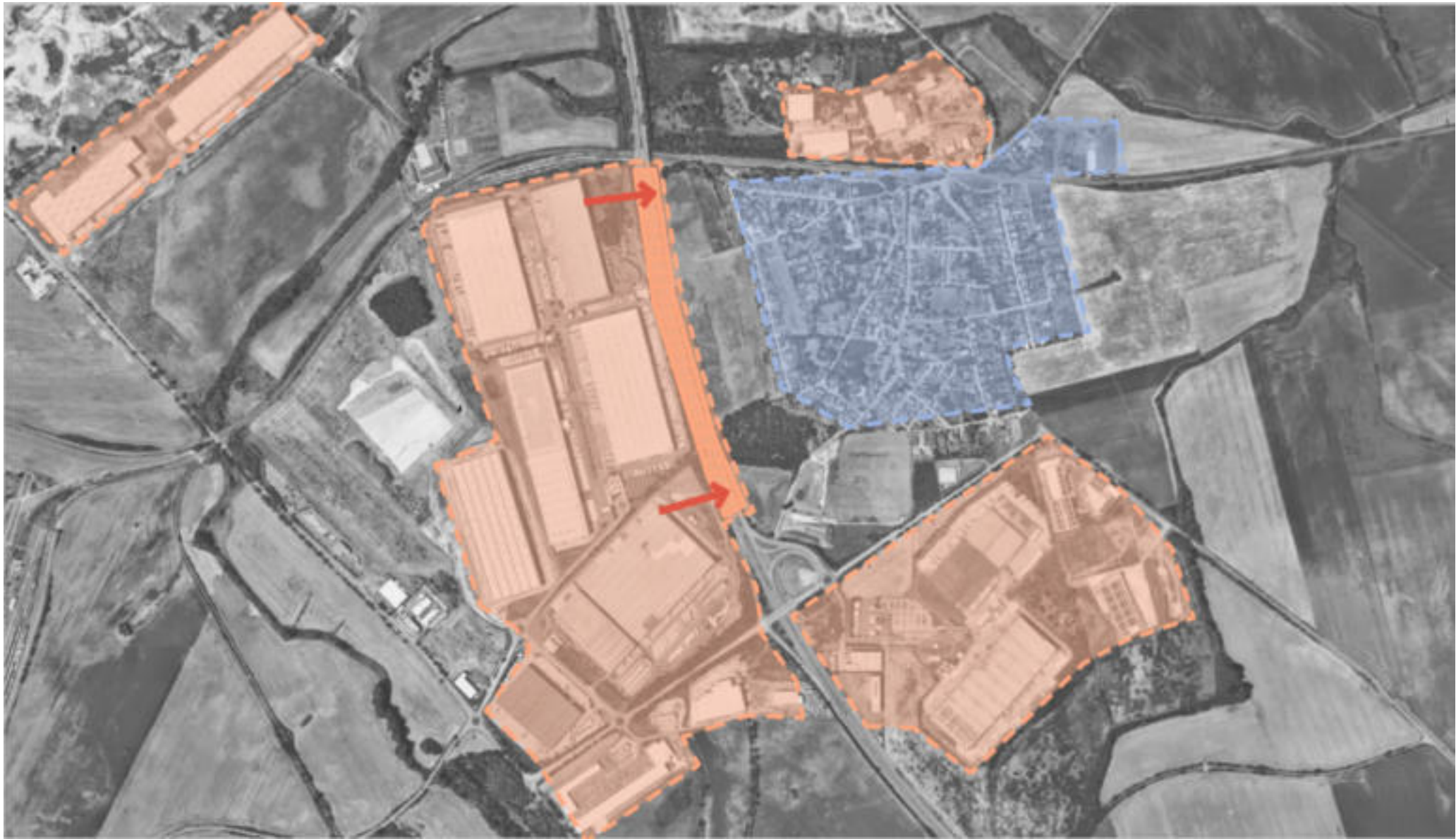
# Other Applications

Since the design is modular, the principle can be applied to any other site where it can be beneficial for different purposes, either for expansion, for removing the large uncomfortable boxes from residential areas, or gain better connectivity.



CTPark Prague Airport, Prague

The warehouses are in very close proximity to the residential area, by implementing this model the residents can gain more access to greenery and a more peaceful environment.



CTPark Prague North, Prague

In this instance, the area could actually gain from the expansion of warehouses, connection people directly to logistics and creating a better connection through the site.



Caravan Metropol, Klíčany-Vodochody

On the last site the logistics is broken up by only one hub on the other side of the highway, by moving it on top, the residents can gain space and have a better connection over the highway.

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